

Programme	B. Sc. STATISTICS				
Course Code	STA 7 CJ 405 (P)				
Course Title	ADVANCED STATISTICAL INFERENCE				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of statistical estimation & testing of hypothesis				
Course Summary	Understand UMVUE and related theorems, UMP tests & SPRT				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe the method of finding sufficient statistics, minimum variance unbiased estimators, consistent estimators and consistent and asymptotically normal estimators	U	F	Instructor-created exams
CO2	Relate sufficient statistic and ancillary statistic using Basu's theorem	Ap	C	Instructor-created exams
CO3	Determine UMVUE using complete sufficient statistic using Rao- Blackwell, and Lehmann-Scheffe theorems	Ap	C	Seminar Presentation
CO4	Explain the concept of interval estimation- SELCI, Bayesian and Fiducial Intervals	U	F	Observation of Practical Skills
CO5	Construct most powerful tests using Neyman-Pearson lemma, one-sided and two-sided UMP tests and UMP unbiased tests	C	M	Group Tutorial Work
CO6	Describe the concept of α -similar tests and construct such tests	U	F	Practical Assignment
CO7	Develop SPRT for different problems	C	P	Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C), Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I		Open-Ended	30	
		Point Estimation. Desirable properties of a good estimator, unbiasedness, efficiency, Methods of Estimation - method of maximum likelihood, Bayesian estimation method. Testing of Hypotheses; concept of testing hypotheses, simple and composite hypotheses, null and alternative hypotheses, type I and type II errors, critical region, level of significance, power of test. Most powerful tests uniformly most powerful test, Neyman Pearson Lemma Problems using R/Python		
II		Sufficient statistics and minimum variance unbiased estimators	12	15
	1	Sufficient statistics, Factorization theorem for sufficiency, Joint sufficient statistics		
	2	Exponential family, Pitman family, Minimal sufficient statistics (MSS). Criteria to find the MSS, Ancillary statistics, Complete statistics		
	3	Basu's theorem		
	4	Unbiasedness, Best Linear Unbiased estimator(BLUE), Minimum variance unbiased estimator(MVUE)		
	5	Rao-Blackwell theorem		
	6	Lehman-Scheffe theorem		
	7	Necessary and sufficient condition for MVUE, Fisher Information, Cramer Rao inequality and its applications		
III		Consistent asymptotically normal estimators and Interval Estimation	12	20
	8	Consistent estimator, Invariance property of consistent estimator		
	9	Method of moments-method of percentiles to determine consistent estimators, choosing between Consistent estimators		
	10	CAN estimators		
	11	Definition of Interval estimation, Shortest expected length confidence interval-large sample confidence intervals-unbiased confidence intervals-examples		
	12	Bayesian and Fiducial intervals		
IV		UMP tests	11	20
	13	One-sided UMP tests, two- sided UMP tests and UMP unbiased tests		
	14	UMP tests for multi-parameter case: UMP unbiased test		
	15	α -similar tests and α -similar tests with Neyman structure, construction of α -similar tests with Neyman structure		
	16	Principle of invariance in testing of hypotheses, locally most powerful tests		
	17	Likelihood ratio tests		
	18	Bayesian tests		
V		Sequential Tests	10	15
	19	Some fundamental ideas of sequential sampling – Sequential Probability Ratio Test (SPRT)		
	20	Important properties, termination of SPRT – the fundamental identity		

		of SPRT		
	21	Operating Characteristic (OC) function and Average Sample Number (ASN) of SPRT		
	22	Developing SPRT for different problems		

Text Books

1. **Kale,B.K . and Muraleedharan K.(2015)** Parametric Inference : An Introduction, Alpha Science Intl Ltd.
2. **George Casella and Roger L Berger (2002).** Statistical inference, Second Edition, Duxbury, Australia.
3. **Manojkumar Srivastava and Namita Srivastava(2009).** Statistical Inference: Testing of Hypothesis, Eastern Economy Edition, PHI Learning Pvt. Ltd., New Delhi.
4. **Rohatgi, V.K(1976).** An introduction to Probability Theory and Mathematical Statistics, John Wiley and sons, New York.

References

1. **Lehmann, E.L(1983).** Theory of point estimation, John Wiley and sons, New York.
2. **Rohatgi, V.K (1984).** Statistical Inference, John Wiley and sons, New York.
3. **Rao, C.R (2002).** Linear Statistical Inference and its applications, Second Edition, JohnWiley and sons, New York.
4. **Lehman, E.L. and Romano, Joseph P.(2005).** Testing Statistical Hypotheses. Third Edition, Springer, New- York.